



Barker College

**2011  
TRIAL  
HIGHER SCHOOL  
CERTIFICATE**

# Mathematics Extension 1

**Staff Involved:**

**AM FRIDAY 12 AUGUST**

- PJR\*    • GIC\*
- MRB    • GDH
- KJL    • RMH
- GPF

**105 copies**

**General Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages of your solutions
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- ALL necessary working should be shown in every question

**Total marks – 84**

- Attempt Questions 1 – 7
- All questions are of equal value
- Marks may be deducted for careless or poorly arranged working

**Total marks – 84**  
**Attempt Questions 1–7**  
**ALL questions are of equal value**

Answer each question on a SEPARATE sheet of paper

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**Marks**

**Question 1** (12 marks) **[START A NEW PAGE]**

- (a) The point  $P(x, y)$  divides the interval  $AB$  internally in the ratio  $2 : 1$  **2**

If  $A$  is the point  $(6, 1)$  and  $B$  is the point  $(12, -8)$ , find the coordinates of  $P(x, y)$

- (b) Evaluate  $\lim_{x \rightarrow 0} \left( \frac{\tan x}{3x} \right)$  **2**

- (c) Use the table of standard integrals to evaluate  $\int_0^{\frac{\pi}{2}} \sec \frac{x}{2} \tan \frac{x}{2} dx$  **2**

- (d) Solve  $\frac{x}{x-4} \leq 2$  **3**

- (e) Evaluate  $\int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{dx}{\sqrt{1-4x^2}}$  **3**

**Question 2** (12 marks) **[START A NEW PAGE]**

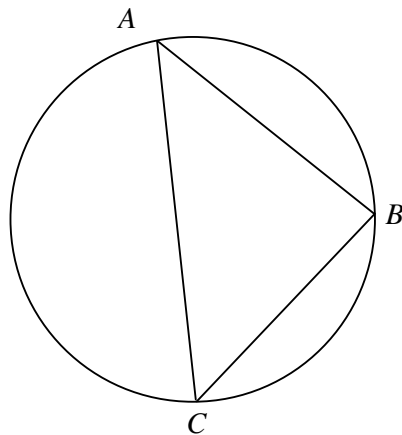
- (a) Find the acute angle between the curves  $y = \log_e x$  and  $y = 1 - x^2$  at the point P (1, 0) 3

Give your answer correct to the nearest minute.

- (b) The point  $P(2ap, ap^2)$  is a point on the parabola  $x^2 = 4ay$  with focus  $S(0, a)$
- (i) Find  $M$ , the midpoint of the chord  $OP$ , where  $O$  is the origin 1
- (ii) Find the gradient of the chord  $OP$  1
- (iii) Find the point  $A$  on the parabola where the tangent is parallel to the chord  $OP$  2
- (iv) Show that  $A$  is equidistant from  $M$  and the  $x$ -axis 1

- (c)  $\triangle ABC$  is inscribed in a circle as shown below.

The tangent at  $C$  meets  $AB$  produced at  $P$  and the bisector of  $\angle ACB$  meets  $AB$  at  $Q$



- (i) Copy and complete the diagram 1
- (ii) Prove that  $PC = PQ$  3

**Question 3** (12 marks) **[START A NEW PAGE]**

- (a) Let  $f(x) = \ln(\tan x)$ , where  $0 < x < \frac{\pi}{2}$  **3**  
 Show that  $f'(x) = 2 \operatorname{cosec} 2x$

- (b) Use the substitution  $x = 2 \sin \theta$  to evaluate  $\int_0^1 \sqrt{4 - x^2} \, dx$  **3**

- (c) (i) State the domain and range of the function  $f(x) = \cos^{-1} 2x$  **2**

- (ii) Draw a neat sketch of the function  $f(x) = \cos^{-1} 2x$  **1**  
 Clearly label all essential features

- (iii) Find the equation of the tangent to the curve  $f(x) = \cos^{-1} 2x$  at the **3**  
 point where the curve crosses the y-axis.

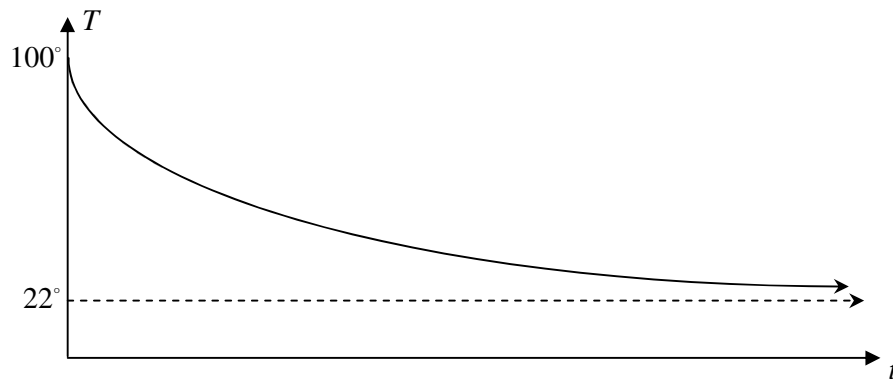
**Question 4** (12 marks) **[START A NEW PAGE]**

(a) (i) Show that  $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$  1

(ii) Hence, or otherwise, evaluate  $\int_0^{\pi/4} \sin 4x \cos 2x \, dx$  3

(b) If  $f(x + 2) = x^2 + 2$ , find  $f(x)$  2

(c) The graph shown below represents the relationship between  $T$ , the temperature in  $^{\circ}\text{C}$  of a cooling cup of coffee, and  $t$ , the time in minutes.



The rate of cooling of this coffee is given by  $\frac{dT}{dt} = -k(T - A)$ , where  $k$  and  $A$  are constants and  $k > 0$ .

(i) Show that  $T = A + Be^{-kt}$  is a solution to the differential equation  $\frac{dT}{dt} = -k(T - A)$ , given that  $B$  is a constant. 1

(ii) By examining the graph when  $t = 0$  and  $t \rightarrow \infty$ , find the values of  $A$  and  $B$  2

(iii) If the temperature of the coffee is  $50^{\circ}\text{C}$  after 90 minutes, show that  $k = -\frac{1}{90} \ln\left(\frac{14}{39}\right)$  2

(iv) Hence, find the rate at which the coffee is cooling after 90 minutes. 1  
Give your answer correct to two significant figures.

**Question 5** (12 marks) **[START A NEW PAGE]**

- (a) Evaluate  $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$  **2**
- (b) The volume of a sphere is increasing at the rate of  $5 \text{ cm}^3$  per second. **3**  
At what rate is the surface area increasing when the radius is  $20 \text{ cm}$  ?
- (c) A particle moves in such a way that its displacement  $x$  cm from an origin  $O$  at any time  $t$  seconds is given by the function  $x = 4 + \sqrt{3} \cos 3t - \sin 3t$
- (i) Show that the particle is moving in simple harmonic motion. **2**
- (ii) Express  $\sqrt{3} \cos 3t - \sin 3t$  in the form  $R \cos(3t + \alpha)$ , where  $\alpha$  is acute and in radians. **2**
- (iii) Find the amplitude of the motion. **1**
- (iv) Find when the particle first passes through the centre of motion. **2**

**Question 6** (12 marks) **[START A NEW PAGE]**

(a) Show by induction that  $7^n + 2$  is divisible by 3, for all positive integers  $n$  **3**

(b) Given the function  $f(x) = \frac{2x+1}{x-1}$

(i) Find any vertical and horizontal asymptotes **1**

(ii) State the domain of the inverse function  $f^{-1}(x)$  **1**

(iii) Sketch the graph of the inverse function  $f^{-1}(x)$  **2**  
Clearly label all critical features of the inverse function  $f^{-1}(x)$

(c) A particle is moving along the  $x$ -axis so that its acceleration after  $t$  seconds is given by

$$\ddot{x} = -e^{-\frac{x}{2}}$$

The particle starts at the origin with an initial velocity of  $2 \text{ cm/sec}$

(i) If  $v$  is the velocity of the particle, find  $v^2$  as a function of  $x$  **2**

(ii) Show that the displacement  $x$  as a function of time  $t$  is given by **3**

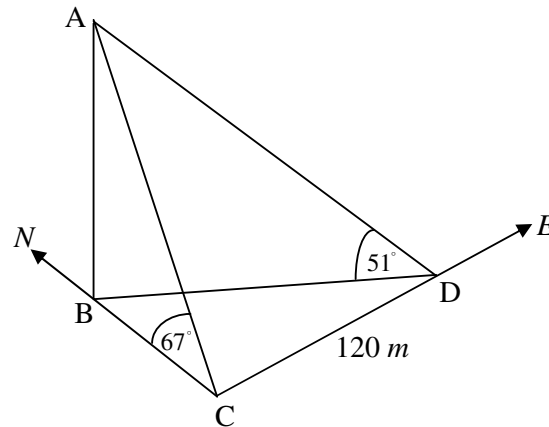
$$x = 4 \log_e \left( \frac{t+2}{2} \right)$$

**Question 7** (12 marks) **[START A NEW PAGE]**

- (a) James is standing at the top A of a tower AB which is built on level ground.

From point C, due south of the base B of the tower, the angle of elevation of the top A of the tower is  $67^\circ$

From point D, which is  $120\text{ m}$  due east of point C, the angle of elevation of the top A of the tower is  $51^\circ$



- (i) Calculate the height of the tower AB (to the nearest metre) **3**

- (ii) James projects a stone horizontally from the top of the tower with velocity  $V\text{ m/s}$

If this stone lands at point D, find the value of  $V$  **3**

(Give your answer correct to one decimal place)

You may assume the equations of motion are

$$x = vt \cos \theta \text{ and } y = vt \sin \theta - 5t^2 \text{ (Do **NOT** prove this)}$$

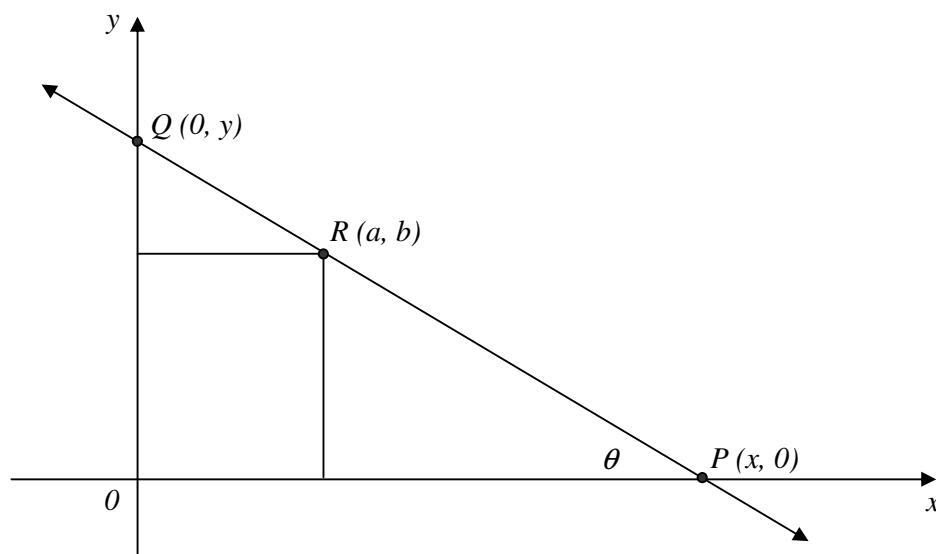
(Hint: Use point A as the origin)

**Question 7 continues on page 9**

Question 7 (continued)

- (b) The point  $R(a, b)$  lies in the positive quadrant of the number plane.

A line through  $R$  meets the positive  $x$  and  $y$  axes at  $P$  and  $Q$  respectively and makes an angle  $\theta$  with the  $x$ -axis.



- (i) Show that the length of PQ is equal to  $\frac{a}{\cos \theta} + \frac{b}{\sin \theta}$  2
- (ii) Hence, show that the minimum length of PQ is equal to  $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{3}{2}}$  4

**End of Question 7**

**End of Paper**

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## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

# AR 12 MATHEMATICS EXTENSION 1 TRIAL HSC

20 August 2011

## Question 1

$$m:n = 2:1$$

$$= \left( \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left( \frac{2 \times 2 + 1 \times 6}{3}, \frac{2 \times -8 + 1 \times 1}{3} \right)$$

$$= \left( \frac{24+6}{3}, \frac{-16+1}{3} \right)$$

$$= (10, -5)$$

$$\frac{1}{3} \lim_{x \rightarrow 0} \frac{\tan x}{x} = \frac{1}{3} \times 1 = \frac{1}{3}$$

$$\left[ 2 \sec \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

$$2 \left( \sec \frac{\pi}{4} - \sec 0 \right)$$

$$2 \times (\sqrt{2} - 1)$$

$$= 2(\sqrt{2} - 1)$$

1st method:

multiplying both sides by the square of the denominator

$$(x-4)^2 \times \frac{x}{x-4} \leq 2(x-4)^2$$

$$(x-4)x \leq 2(x^2 - 8x + 16)$$

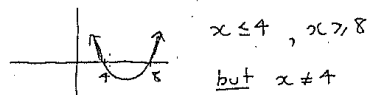
$$x^2 - 4x \leq 2x^2 - 16x + 32$$

$$0 \leq x^2 - 12x + 32$$

$$x^2 - 12x + 32 \geq 0$$

$$(x-4)(x-8) \geq 0$$

zeros are  $x=4, x=8$



because it is a zero of the denominator

$$\therefore x < 4 \text{ or } x > 8$$

2nd Method: graphical

$$\text{for } \frac{x}{x-4} \leq 2$$

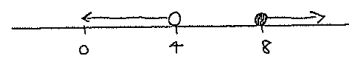
$$1. x \neq 4$$

$$2. \text{ solve } \frac{x}{x-4} = 2$$

$$x = 2x - 8$$

$$x = 8$$

$\therefore$  critical points are  $x=4, 8$



check all 3 regions:

$$i) x < 4, x=0, \frac{0}{0-4} \leq 2$$

true  $\therefore x < 4$

$$ii) 4 < x < 8, x=5, \frac{5}{5-4} \leq 2$$

not true

$$iii) x > 8, x=9, \frac{9}{9-4} \leq 2$$

$$\therefore x < 4 \text{ or } x > 8$$

$$e) \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{dx}{\sqrt{1-(2x)^2}}$$

$$= \frac{1}{2} \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{2 dx}{\sqrt{1-(2x)^2}}$$

$$= \frac{1}{2} \left[ \sin^{-1} 2x \right]_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= \frac{1}{2} \left( \sin^{-1} \frac{1}{2} - \sin^{-1} \left( -\frac{1}{2} \right) \right)$$

$$= \frac{1}{2} \left( \sin^{-1} \frac{1}{2} + \sin^{-1} \frac{1}{2} \right)$$

$$= \frac{1}{2} \times 2 \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{6}$$

## Question 2

$$a) \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$y_1 = \log_e x$$

$$y_2 = 1 - x^2$$

$$y'_1 = \frac{1}{x}$$

$$y'_2 = -2x$$

$$\text{at } x=1, m_1 = y'_1 = \frac{1}{1} = 1$$

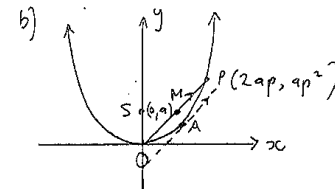
$$x=1, m_2 = y'_2 = -2 \times 1 = -2$$

$$\therefore \tan \theta = \left| \frac{1 - (-2)}{1 + (-2)} \right|$$

$$= \left| \frac{3}{-1} \right|$$

$$\tan \theta = 3$$

$$\theta = \tan^{-1} 3 = 71^\circ 34'$$



$$i) M = \left( \frac{0+2ap}{2}, \frac{0+ap^2}{2} \right)$$

$$M = \left( ap, \frac{ap^2}{2} \right)$$

$$ii) \text{ Gradient of OP} = \frac{ap^2 - 0}{2ap - 0}$$

$$\therefore m_{OP} = \frac{p}{2}$$

iii) A lies on the parabola

$$x^2 = 4ay \therefore y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a} = \frac{x}{2a}$$

now  $y' = \frac{p}{2}$  since chords are parallel

$$\frac{x}{2a} = \frac{p}{2}$$

$$x = ap$$

$$\text{if } x = ap, y = \frac{(ap)^2}{4a} = \frac{ap^2}{4}$$

$$\therefore A \left( ap, \frac{ap^2}{4} \right)$$

iv) Distance of A from x-axis

= its y value

$$\therefore d_1 = \frac{ap^2}{4} \text{ units}$$



$$\begin{aligned} & \frac{1}{2} \int_0^{\frac{\pi}{4}} \sin 6x + \sin 2x \, dx \\ & \frac{1}{2} \left[ \frac{-\cos 6x}{6} - \frac{\cos 2x}{2} \right]_0^{\frac{\pi}{4}} \\ & = -\frac{1}{2} \left( \frac{\cos \frac{3\pi}{2}}{6} + \frac{\cos \frac{\pi}{2}}{2} - \left( \frac{\cos 0}{6} + \frac{\cos 0}{2} \right) \right) \\ & = -\frac{1}{2} \left( 0 + 0 - \left( \frac{1}{6} + \frac{1}{2} \right) \right) \\ & = -\frac{1}{2} \times -\frac{2}{3} \\ & = \frac{1}{3} \end{aligned}$$

1) 1st method:

$$\begin{aligned} f(x+2) &= (x+2)^2 - 4x - 2 \\ & \quad \uparrow \\ & \text{rewrite RHS in terms} \\ & \text{of } (x+2)^2 \end{aligned}$$

$$f(x+2) = (x+2)^2 - 4(x+2) + 6 \quad \therefore x+2 = (x+2)^2 - 4(x+2) + 6$$

$$\therefore f(x) = x^2 - 4x + 6$$

(replace  $(x+2)$  with  $x$ )

2nd method:

$$\begin{aligned} f(x) &= f((x-2)+2) \\ &= (x-2)^2 + 2 \\ &= x^2 - 4x + 4 + 2 \\ &= x^2 - 4x + 6 \end{aligned}$$

3rd method:

$$\text{Equate } x^2 + 2 \equiv A(x+2)^2 + B(x+2) + C$$

sub  $x = -2$

$$6 \equiv 0 + 0 + C$$

$$C = 6$$

$$\text{sub } x = 0, \quad 2 = 4A + 2B + C$$

$$2 = 4A + 2B + 6$$

$$2A + B = -2$$

$$\text{sub } x = -1, \quad 3 = A + B + C$$

$$A + B = -3$$

$\therefore$  solve simultaneously

$$\begin{cases} 2A + B = -2 & \text{①} \\ A + B = -3 & \text{②} \end{cases}$$

$$\text{①} - \text{②} \quad A = 1$$

$$\therefore B = -4$$

$$\therefore f(x+2) = (x+2)^2 - 4(x+2) + 6$$

$$f(x) = x^2 - 4x + 6$$

c) 1) Method 1

$$T = A + B e^{-kt}$$

$$\frac{dT}{dt} = -k B e^{-kt}$$

$$= -k(A + B e^{-kt} - A)$$

$$= -k(T - A)$$

$$\therefore \frac{dT}{dt} = -k(T - A)$$

Method 2

$$T = A + B e^{-kt}$$

$$\text{show that } \frac{dT}{dt} = -k(T - A)$$

$$\text{LHS} = \frac{dT}{dt} = -k B e^{-kt}$$

$$\text{RHS} = -k(A + B e^{-kt} - A)$$

$$= -k(B e^{-kt})$$

$$= -k B e^{-kt}$$

$$= \text{LHS}$$

$$\text{ii) from graph } t=0, T=100^\circ$$

$$\text{sub } t=0, \quad 100 = A + B e^0$$

$$A + B = 100$$

$$t \rightarrow \infty, T \rightarrow 22^\circ$$

$$\text{as } t \rightarrow \infty \quad B e^{-kt} \rightarrow 0$$

$$22 = A + 0$$

$$\therefore A = 22$$

$$\text{and } B = 78$$

$$\therefore T = 22 + 78 e^{-kt}$$

$$\text{iii) } T = 50, t = 90$$

$$50 = 22 + 78 e^{-90k}$$

$$28 = 78 e^{-90k}$$

$$e^{-90k} = \frac{14}{39}$$

$$-90k = \log_e \left( \frac{14}{39} \right)$$

$$k = -\frac{1}{90} \log_e \left( \frac{14}{39} \right)$$

iv) Rate is

$$\frac{dT}{dt} = -k(T - A)$$

$$\frac{dT}{dt} = \frac{1}{90} \log_e \left( \frac{14}{39} \right) (50 - 22)$$

$$= \frac{1}{90} \log_e \left( \frac{14}{39} \right) \times 28$$

$$= -0.318 T \dots$$

$$= -0.32 \text{ } ^\circ\text{C/min}$$

(to 2 sig figs)

Question 5

$$a) \left[ \frac{5\omega^3 x}{3} \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} (5\omega^3 \frac{\pi}{4} - 5\omega^3 0)$$

$$= \frac{1}{3} \times \left( \frac{1}{\sqrt{2}} \right)^3$$

$$= \frac{1}{3} \times \frac{1}{2\sqrt{2}}$$

$$= \frac{1}{6\sqrt{2}} \quad (\text{or } \frac{\sqrt{2}}{12})$$

$$b) \frac{dV}{dt} = 5 \text{ cm}^3/\text{s}$$

$$\text{now } \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\therefore 5 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{5}{4\pi r^2}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = 8\pi r \times \frac{dr}{dt}$$

$$= 8\pi \times \frac{5}{4\pi r}$$

$$= \frac{10}{r}$$

when  $r = 20$  cm

$$\frac{dA}{dt} = \frac{10}{20} = \frac{1}{2}$$

∴ rate at which surface area is increasing is  $\frac{1}{2} \text{ cm}^2/\text{s}$

$$c) i) x = 4 + \sqrt{3} \cos 3t - \sin 3t$$

$$\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$\ddot{x} = -9(\sqrt{3} \cos 3t - \sin 3t + 4) + 36$$

$$\ddot{x} = -9x + 36$$

$$= -9(x-4)$$

which is SHM,  $n=3$ , centre is 4 cm

$$\sqrt{3} \cos 3t - \sin 3t = R \cos 3t \cos \alpha - R \sin 3t \sin \alpha$$

Equating both sides:

$$\left. \begin{aligned} R \cos \alpha &= \sqrt{3} \\ R \sin \alpha &= 1 \end{aligned} \right\} \begin{aligned} (1) \\ (2) \end{aligned}$$

$$(2) \div (1) \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$R = \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1^2} = \sqrt{4} = 2$$

$$\therefore 2 \cos\left(3t + \frac{\pi}{6}\right)$$

$$iii) \text{ so, } x = 4 + 2 \cos\left(3t + \frac{\pi}{6}\right)$$

$$\text{since } -1 \leq \cos\left(3t + \frac{\pi}{6}\right) \leq 1$$

then  $x$  can be between  $(4+2)$  cm and  $(4-2)$  cm

$$\text{i.e. } 2 \leq x \leq 6$$

the centre is  $x=4$

∴ amplitude is 2 cm

iv) solve:

$$4 + 2 \cos\left(3t + \frac{\pi}{6}\right) = 4$$

$$2 \cos\left(3t + \frac{\pi}{6}\right) = 0$$

$$\cos\left(3t + \frac{\pi}{6}\right) = 0$$

$$3t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

1st time

$$\therefore 3t + \frac{\pi}{6} = \frac{\pi}{2}$$

$$3t = \frac{\pi}{3}$$

$$t = \frac{\pi}{9}$$

∴ particle first passes

through  $x=4$  after  $t = \frac{\pi}{9}$  sec

### Question 6

$$a) \text{ show that } 7^n + 2 = 3N$$

where  $N$  and  $n$  are integers both  $> 1$

Prove true for  $n=1$

$$\text{LHS} = 7^1 + 2 = 9 = 3 \times 3$$

$$= \text{RHS for } N=3$$

∴ statement true for  $n=1$

Assume true for  $n=k$  where

$k$  is an integer  $> 1$

$$\text{i.e. } 7^k + 2 = 3N$$

now prove true for  $n=k+1$

$$7^{k+1} + 2 = 3M \text{ where } M \text{ is a positive integer}$$

$$\text{LHS} = 7^{k+1} + 2$$

$$= 7 \times 7^k + 2$$

$$= 7 \times (3N - 2) + 2 \quad (\text{from assumption})$$

$$= 21N - 14 + 2$$

$$= 21N - 12$$

$$= 3(7N - 4)$$

$$= 3M \text{ where } M = 7N - 4$$

$$= \text{RHS}$$

∴ statement is true for  $n=k+1$

∴ statement is true for  $n=1$ ,

$n=k$  and  $n=k+1$

∴ it is true for all positive integers  $n$

$$b) i) f(x) = \frac{2x+1}{x-1}$$

vertical asymptote  $x=1$

(as  $x-1 \neq 0$ )

horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{2x+1}{x-1} = \lim_{x \rightarrow \infty} \frac{\frac{2x}{x} + \frac{1}{x}}{\frac{x}{x} - \frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{2 + \frac{1}{x}}{1 - \frac{1}{x}}$$

$$= \frac{2+0}{1-0}$$

$$= 2$$

∴  $y=2$  is horiz. asymptote

ii) Domain of  $f^{-1}(x)$  is the same as range of  $f(x)$   
∴  $x \neq 2$

iii) Sketch  $y = f^{-1}(x)$

$$D: x \neq 2$$

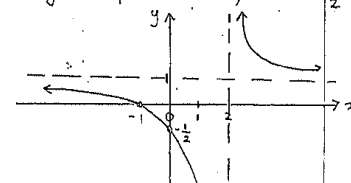
$$R: y \neq 1$$

for  $y=f(x)$  the  $y$ -intercept is -1

∴  $x$  intercept of  $f^{-1}(x)$  is -1

for  $y=f(x)$  the  $x$ -intercept is  $-\frac{1}{2}$

∴  $y$  intercept of  $f^{-1}(x)$  is  $-\frac{1}{2}$



$$\begin{aligned}
 i) \quad \frac{1}{2} V^2 &= \int -e^{-\frac{x}{2}} dx \\
 \frac{1}{2} V^2 &= -\frac{e^{-\frac{x}{2}}}{-\frac{1}{2}} + C \\
 \frac{1}{2} V^2 &= 2e^{-\frac{x}{2}} + C
 \end{aligned}$$

$$= 0, v = 2$$

$$2 = 2 + C$$

$$\therefore C = 0$$

$$\therefore V^2 = 4e^{-\frac{x}{2}}$$

$$V = \pm \sqrt{4e^{-\frac{x}{2}}}$$

$$V = \pm 2e^{-\frac{x}{4}}$$

$$\text{at } x=0, V=2 \therefore \text{take}$$

positive  $V$

$$V = 2e^{-\frac{x}{4}}$$

$$\frac{dx}{dt} = 2e^{-\frac{x}{4}} = \frac{2}{e^{\frac{x}{4}}}$$

$$\therefore \int \frac{e^{\frac{x}{4}}}{2} dx = \int dt$$

$$\frac{1}{2} \left( \frac{e^{\frac{x}{4}}}{\frac{1}{4}} \right) = t + K$$

$$\frac{1}{2} \times 4e^{\frac{x}{4}} = t + K$$

$$2e^{\frac{x}{4}} = t + K$$

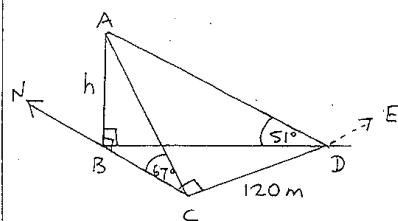
$$t=0, x=0 \quad 2 = 0 + K$$

$$\therefore K = 2$$

$$2e^{\frac{x}{4}} = t + 2$$

$$\begin{aligned}
 e^{\frac{x}{4}} &= \frac{t+2}{2} \\
 \frac{x}{4} &= \log_e \left( \frac{t+2}{2} \right) \\
 \therefore x &= 4 \log_e \left( \frac{t+2}{2} \right)
 \end{aligned}$$

### Question 7



$$i) \tan 67^\circ = \frac{h}{BC} \quad \tan 51^\circ = \frac{h}{BD}$$

$$BC = \frac{h}{\tan 67^\circ} \quad BD = \frac{h}{\tan 51^\circ}$$

$$BC = h \tan 23^\circ \quad BD = h \tan 39^\circ$$

In the base  $\triangle BDC$ :

use Pythagoras' Theorem since

$$\angle BCD = 90^\circ$$

$$BD^2 = BC^2 + 120^2$$

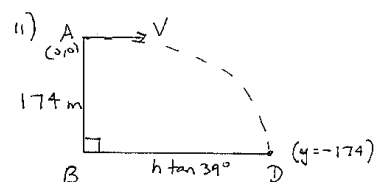
$$(h \tan 39^\circ)^2 = (h \tan 23^\circ)^2 + 120^2$$

$$h^2 (\tan^2 39^\circ - \tan^2 23^\circ) = 120^2$$

$$h^2 = \frac{120^2}{\tan^2 39^\circ - \tan^2 23^\circ}$$

$$h = \frac{120}{\sqrt{\tan^2 39^\circ - \tan^2 23^\circ}}$$

$$\therefore h = 174 \text{ m (nearest m)}$$



horizontal projection  $\therefore \theta = 0$

$$x = Vt \cos \theta, \quad y = Vt \sin \theta - 5t^2$$

$$\theta = 0$$

$$x = Vt, \quad y = -5t^2$$

$$\text{At point D, } y = -174$$

$$-174 = -5t^2$$

$$t^2 = \frac{174}{5}$$

$$t = \sqrt{\frac{174}{5}} = 5.9 \text{ sec}$$

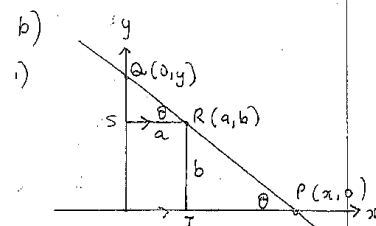
then  $x = Vt$

$$h \tan 39^\circ = V \times 5.9$$

$$V = \frac{174 \times \tan 39^\circ}{5.9}$$

$$= 23.8817 \dots$$

$$\therefore V = 23.9 \text{ m/s}$$



$$i) \text{ length } PA = QR + RP$$

$$\text{In } \triangle RTP: \sin \theta = \frac{b}{RP}$$

$$\therefore RP = \frac{b}{\sin \theta}$$

$$\text{In } \triangle QRS: \cos \theta = \frac{a}{RQ}$$

$$RQ = \frac{a}{\cos \theta}$$

$$\therefore PA = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$ii) \text{ let } L = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$\text{ie } L = a \sec \theta + b \csc \theta$$

to minimise length, solve

$$\frac{dL}{d\theta} = 0 \text{ for } \theta$$

$$L' = a \sec \theta \tan \theta + -b \csc \theta \cot \theta$$

$$= a \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta} - b \times \frac{1}{\sin \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$= \frac{a \sin \theta}{\cos^2 \theta} - \frac{b \cos \theta}{\sin^2 \theta} = 0 \text{ for min}$$

$$\therefore \frac{a \sin \theta}{\cos^2 \theta} = \frac{b \cos \theta}{\sin^2 \theta}$$

$$a \sin^3 \theta = b \cos^3 \theta$$

$$\therefore \tan^3 \theta = \frac{b}{a}$$

$$\tan \theta = \sqrt[3]{\frac{b}{a}} = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}$$

$$\theta = \tan^{-1}\left(\sqrt[3]{\frac{b}{a}}\right)$$

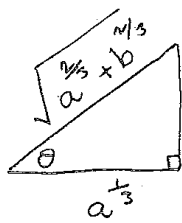
$\therefore \theta = \tan^{-1}\left(\sqrt[3]{\frac{b}{a}}\right)$  gives the

minimum length of PQ

(\* it is a min. length since there is no max. value for

PQ, since as  $\theta \rightarrow 0$ , length PQ  $\rightarrow \infty$ )

now, since  $\tan \theta = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}$



use Pythagoras

$$\therefore \sin \theta = \frac{b^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$$\cos \theta = \frac{a^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$\therefore$  Minimum length of PQ:

$$PQ = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$= \frac{a}{\frac{a^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}} + \frac{b}{\frac{b^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}}$$

$$= \frac{a^{\frac{2}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}} + \frac{b^{\frac{2}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$$PQ = a^{\frac{2}{3}} \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}} + b^{\frac{2}{3}} \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}$$

$$= \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}} \left( a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)$$

$$= \left( a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{1}{2}} \left( a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)$$

$$= \left( a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$$

!! when!

$\therefore$  min length of PQ is

equal to  $\left( a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$